

A. Physical Read Overlap Caliper Model (PE150)

Physical Centromeric Fragment: $L = 133$ bp

P7 Adapter
(17 bp)

Read 1: Insert (133 bp)

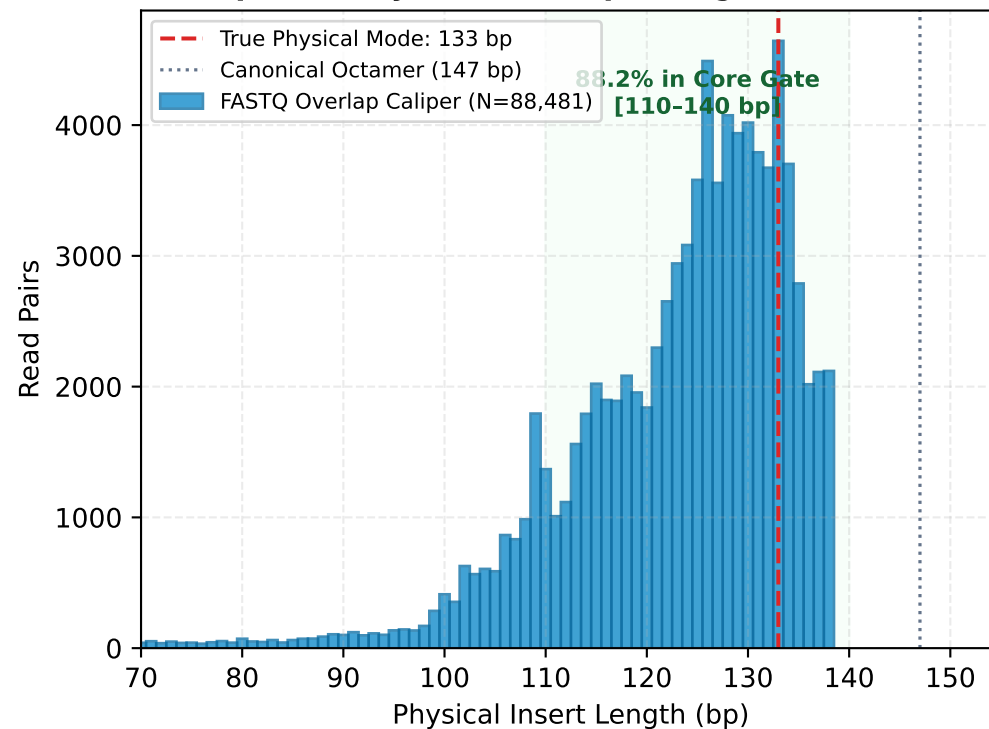
P5 Adapter
(17 bp)

Read 2: Rev Comp Insert (133 bp)

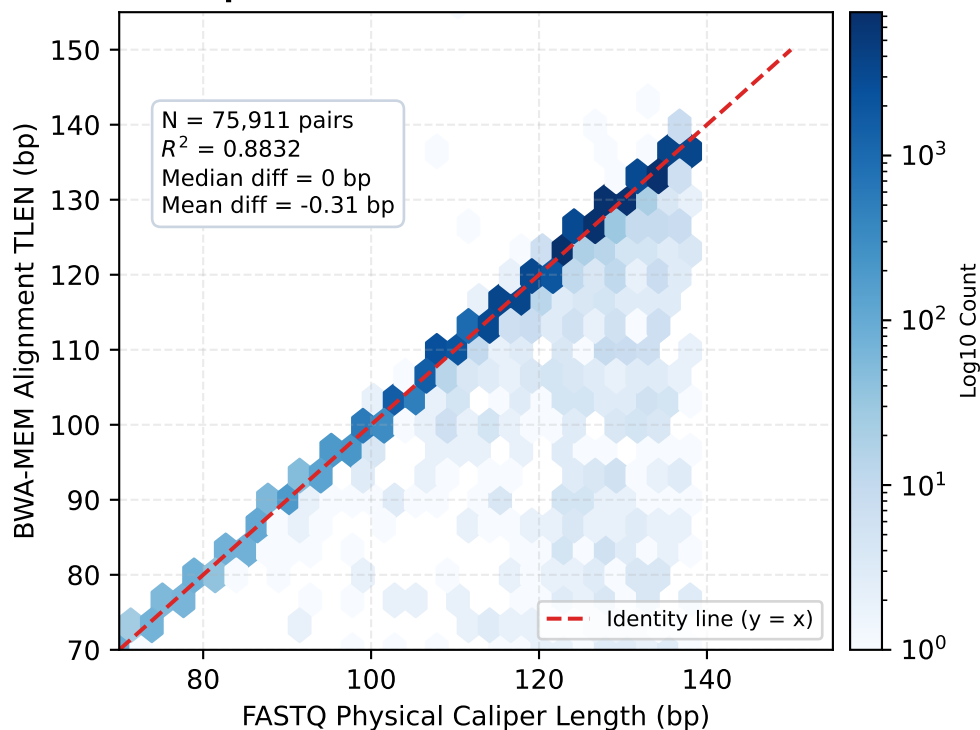
133-bp Full Physical Duplex Overlap

- Bounded by 3' adapter start: $L = p1 = p2$
- Mates agree base-for-base (0 true mismatches)
- Independent of reference genome & aligner!

B. Empirical Physical Overlap Sizing (Raw FASTQ)



C. Caliper vs BAM TLEN Concordance



D. Invariance of Open Core Mode Across MAPQ Strata

